CDI COLLEGE | BDA400

Assignment 6

Technical Analysis Using R

Visualization Stage | Portfolio Dashboard

Student and Course

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Course	CA-BDA400 - Data Science Tools and Techniques
Assignment	Assignment 6: Technical Analysis Using R, Visualization Stage

Project Overview

This project extends the earlier technical-analysis work into an interactive R Shiny portfolio dashboard. It retrieves historical stock prices from Yahoo Finance and lets the user explore selected stocks across date ranges and daily, weekly, or monthly intervals. Price data can be displayed as line, area, or candlestick charts.

Dashboard Features

- Selectable AAPL, MSFT, and GOOGL stock symbols with an adjustable date range and time frame.
- Toggleable 20-day and 50-day simple moving averages, RSI (14), and MACD indicators.
- Moving-average crossover rules that generate Buy, Sell, or Hold signals, with Buy and Sell annotations on the price chart.

Public Project Repository

https://github.com/maljarad30-coder/TechnicalAnalysis

Prepared for submission. The R Shiny source code is maintained in the public GitHub repository.